

European Research Council

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QUANTUM-CELL

Non-Linear Quantum Coherence in Single-Cell Epigenomic State Dynamics

Primary Domain: PE3 Condensed Matter Physics | Secondary Panel: LS2 Genetics & Genomics

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Principal Investigator:	Dr. Elena Ros-tova	Call Identifier:	ERC-2026-StG
Host Institution:	Vertex Institute for Advanced Science	Proposal Duration:	60 Months
Collaborating Partner:	Synapse Quantum Biosystems Center	Total Requested Budget:	€1,498,500

Part B1 – Section a: Extended Synopsis

1.1 Executive Summary & Groundbreaking Nature

Classical molecular biology treats single-cell state transitions as purely stochastic chemical reactions governed by thermal fluctuations. However, recent unexpected discoveries in high-resolution ultrafast optical spectroscopy suggest that non-equilibrium quantum-coherent phenomena may persist in biological macromolecules at ambient temperatures (300 K). The **QUANTUM-CELL** project aims to establish a radical new paradigm by proving that sub-picosecond quantum coherence within chromatin-bound protein complexes directly modulates non-linear gene expression trajectories during stem cell differentiation.

By engineering a novel hybrid platform integrating *ultrafast single-cell optical resonance spectroscopy* with *in situ high-throughput spatial epigenomics*, QUANTUM-CELL will map quantum state manifolds in single living cells for the first time. If successful, this research will rewrite fundamental assumptions in bioenergetics and open unprecedented therapeutic avenues for targeted cell reprogramming.

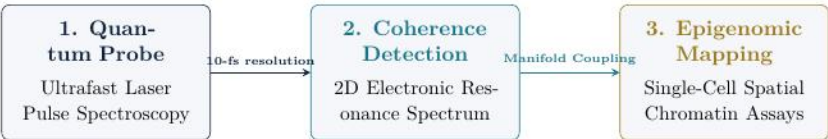


Figure 1: Concept overview of the QUANTUM-CELL experimental pipeline linking ultrafast optical physics to single-cell genomics.